



Giuseppe Mattia Greco

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[REDACTED]

EDUCATION AND TRAINING

09/2024 – CURRENT Arcavacata di Rende (CS), Italy
MASTER'S DEGREE IN COMPUTER ENGINEERING University of Calabria

- Ethical Hacking
- Software and Data Security
- Network Security
- IoT Security
- Cryptography
- Artificial Intelligence basics

Address Via Pietro Bucci, snc, 87036, Arcavacata di Rende (CS), Italy | **Website** <https://www.unical.it/storage/cds/47759/> |

Field of study Cybersecurity

10/2017 – 07/2024 Arcavacata di Rende (CS), Italy
BACHELOR'S DEGREE IN COMPUTER ENGINEERING University of Calabria

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Field of study Computer Engineering |

Thesis Applicazione di tecniche di Machine Learning per la diagnosi di malattie cardiache da dati raccolti da dispositivi indossabili

10/09/2012 – 08/06/2017 Polistena (RC), Italy
IT EXPERT Istituto Tecnico Industriale Statale "M.M. Milano"

Address Via dello Sport 25, 89024, Polistena (RC), Italy | **Website** <https://www.itispolistena.edu.it/> | **Field of study** Computer Science

WORK EXPERIENCE

SOFTWARE ENGINEERING INTERN - IT-WALLET – DEPARTMENT FOR DIGITAL TRANSFORMATION (PRESIDENCY OF THE COUNCIL OF MINISTERS) – 26/11/2025 – 26/02/2026 – ROME (RM), ITALY

Completed a curricular internship at the Department for Digital Transformation (Presidency of the Council of Ministers), focusing on the architectural analysis and Quality Assurance of the Italian IT-Wallet infrastructure in compliance with the eIDAS 2.0 framework.

Key Responsibilities & Achievements:

- **Quality Assurance & Gap Analysis:** conducted rigorous gap analysis on system Test Plans to identify and resolve vulnerabilities in the Credential Issuer node low-level flows;
- **Official Specification Contribution:** designed and integrated test case CI_061a to enforce PKCE security (code_verifier), which was officially merged into the government repository and included in Release 1.4.0;
- **Identity Proxy Testing:** Performed functional and integration testing on the iam-proxy-italia component, validating interactions between SAML/OIDC frontends and OpenID4VP backends;
- **European Open Source Collaboration:** contributed to the WE BUILD Consortium (WP4 - Trust Group) by refactoring the Trust Infrastructure specifications, successfully merging contributions via GitHub.

Technical Skills:

- **Identity & Access Management:** deep understanding of eIDAS 2.0, OpenID4VCI, OpenID4VP, SAML2, and OIDC protocols;
- **Software Testing:** independent QA, network/API testing, and deterministic testing methodologies;
- **Version Control & CI/CD:** professional use of Git, GitHub (Pull Requests, Peer Review), and Continuous Integration pipelines.

Technologies utilized: eIDAS 2.0, OpenID4VCI, OpenID4VP, SAML2, OIDC, Git, GitHub, Docker, cURL, Markdown, reStructuredText.

● **SKILLS**

Programming and Markup Languages

Python | C/C++ | Java | Bash | LaTeX | JavaScript | Assembly (x86/64) | Markdown | reStructuredText | HTML/CSS | SQL | JSON

Artificial Intelligence & Data Science

Natural Language Processing (NLP) | Data Visualization (NumPy, Pandas, Matplotlib) | Generative AI (LLMs, LangChain, FASS) | MLOps | Machine Learning

Cybersecurity & Identity

Cybersecurity Principles | Experience in CTF | Identity & Access Management (IAM) | SAML2

Software Engineering & DevOps

Software/API Testing | Cloud Systems (Amazon AWS, Microsoft Azure, Google Cloud Platform) | Software Design Patterns | Git | GitHub (CI/CD, Code Review) | IAC with Terraform | Docker

Operative Systems

Linux | Windows | MacOS | Hardware/Software Troubleshooting

Soft Skills

Problem Solving | Work in multidisciplinary teams | Asynchronous Collaboration | Adaptability | Communication

● **LANGUAGE SKILLS**

Mother tongue(s): **ITALIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B1	B1	B1	B1	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **PROJECTS**

05/2024 – 07/2024

Bachelor's Degree Thesis - Detection of heart rate anomalies using the Matrix Profile algorithm

Three-year thesis project focused on the development of an intelligent system in Python for the automatic detection of anomalies in heart rate data. The work implements the Matrix Profile algorithm applied to the analysis of biomedical time series, with the aim of identifying irregular patterns in heartbeats through advanced data science techniques.

Technical skills:

- Machine Learning & Data Science: Implementation of the Matrix Profile algorithm for pattern analysis in time series;
- Data Preprocessing: Normalisation and cleaning of raw biomedical data;
- Data Visualisation: Creation of graphs and visualisations for the interpretation of detected anomalies;
- Python: Use of scientific libraries (NumPy, Pandas, Matplotlib) for quantitative analysis;
- Healthcare Analytics: Application of data science techniques to the medical/healthcare sector.

Technologies used:

Python, Matrix Profile Algorithm, Time Series Analysis, Data Visualisation, Scientific Computing

Link <https://github.com/giumatt/Heartrate-Anomaly-Detection-Matrix-Profiling>

11/2024 – 01/2025

Simulated Annealing for Protein Folding Prediction

Developed an advanced system for predicting protein tertiary structures using the Simulated Annealing algorithm. The project focuses on optimising computational performance through implementation in C, with critical sections optimised in Assembly and parallelisation via OpenMP to maximise computational efficiency.

Key features:

- **Simulated Annealing algorithm:** Implementation of the meta-heuristic algorithm to explore the conformational space of proteins;
- **Low-Level optimizations:** Critical sections of code rewritten in Assembly to maximize performance;
- **Parallelization:** Use of OpenMP to exploit multi-core architectures and reduce computation times;
- **High-Performance Computing:** Focus on computational efficiency and resource optimization.

Technical skills:

- **C Programming:** Development of the application core (85.4% of the project);
- **Assembly:** Optimization of computationally intensive low-level routines;
- **Parallel Computing:** Implementation of parallel algorithms with OpenMP;
- **Performance Optimization:** Code profiling and optimisation to improve efficiency;
- **Team Collaboration:** Collaborative development and teamwork coordination.

Technologies used:

C, Assembly (x86/x64), OpenMP, Shell Scripting, Simulated Annealing Algorithm, High-Performance Computing

Link <https://github.com/giumatt/Simulated-Annealing-for-Protein-Folding-Prediction>

12/2024 – 06/2025

Sentiment Analysis for a Smart Museum

Developed an innovative sentiment analysis system for smart museums that detects and displays visitors' emotional responses to works of art. The project integrates Artificial Intelligence and Natural Language Processing (NLP) techniques to create an interactive, data-driven museum experience, providing valuable insights into audience preferences and reactions.

Key features:

- **Emotion Detection:** Automatic analysis of visitor sentiment through textual feedback and interactions;
- **Data Visualization:** Interactive dashboards to represent aggregated emotional responses per artwork;
- **Multi-language Architecture:** Modular system combining Java (backend), Python (AI/ML) and C++ (performance-critical components);
- **Smart Museum Integration:** Solution designed for integration into digital museum ecosystems;
- **Team Collaboration:** Collaborative development and coordination of teamwork.

Technologies used:

Java, Python, C++, Natural Language Processing, Sentiment Analysis, Machine Learning, Data Visualisation, AI

Link <https://github.com/giumatt/Sentiment-Analysis-for-a-Smart-Museum>

05/2025 – 06/2025

Winalyze

Developed an end-to-end Machine Learning Operations (MLOps) system for predicting wine quality. The project implements a complete pipeline that manages the entire ML model lifecycle, from data upload to production inference, with a modern user interface developed in Vue.js to facilitate interaction with the system.

Key features:

- **Data Upload & Management:** System for uploading and managing wine datasets;
- **Model Training:** Automated pipeline for training machine learning models;
- **Model Validation:** Validation framework to ensure model quality and accuracy;
- **Inference Engine:** Real-time prediction system to evaluate wine quality;
- **Modern Frontend:** Interactive user interface developed in Vue.js for a user-friendly experience;
- **Full-Stack MLOps:** Comprehensive architecture that integrates frontend and backend ML.

Technologies used:

Vue.js, Python, TypeScript, CSS, HTML, Machine Learning, MLOps, RESTful APIs, Data Pipeline, Model Deployment, Microsoft Azure

Link <https://github.com/giumatt/Winalyze>

07/2025 – 02/2026

QuestMasterAI

Developed as a group-based interdisciplinary project, this is a comprehensive system for the automatic generation of interactive narrative quests, integrating Large Language Models (LLM) with formal planning methodologies. The project ensures the logical consistency and solvability of the generated stories through the automatic validation of action plans.

Key features:

- **AI-Driven Planning:** automatic transformation of natural language descriptions into formal PDDL representations (domain and problem);
- **Reflective Agent:** implementation of an intelligent agent capable of analyzing planning (Fast-Downward) and validation (VAL) errors, automatically proposing corrections to PDDL files;
- **RAG Integration:** Retrieval-Augmented Generation system to enrich narrative lore with information extracted from external PDF documents via a vector store;
- **Web Architecture:** interactive user interface based on Flask for Human-in-the-Loop management of the entire

pipeline, from lore creation to final gameplay.

Technical skills:

- Generative AI: advanced use of LLMs for generating formal code, creative content, and iterative reasoning (Chain-of-Thought);
- Automated Planning: modeling complex domains and problems using the PDDL standard and integrating domain-independent planners;
- Software Orchestration: development of modular pipelines for the integration of AI components, vector databases, and external tools.

Technologies used: Python, LangChain, Google Gemini 2.5 Flash, PDDL (Fast-Downward, VAL), FAISS, Flask, PyPDF.

01/2026 – 06/2026

Smart MicroGrid

Developed as a group course project, this secure gateway for Smart MicroGrids focuses on the integration of advanced authentication mechanisms, encryption, and immutable traceability of operational data. The system ensures end-to-end security and the integrity of communications between IoT devices and the smart grid infrastructure.

Key features:

- Encrypted Communication (mTLS): implementation of mutual TLS authentication (mTLS) to secure communication channels and verify the digital identity of microgrid nodes communicating with the gateway;
- IoT Trust Engine: integration of the MQTT protocol for lightweight message exchange, supported by a “Trust Engine” for dynamic risk assessment and device access control;
- Blockchain Notarization: immutable tracking and logging of telemetry data and critical network events using blockchain technology, ensuring the integrity of historical data and non-repudiation;
- Containerized Architecture: virtualization and isolation of gateway services via Docker to facilitate the deployment and orchestration of security components.

Technical Skills:

- IoT Security: application of hardening principles, applied cryptography, and credential management in embedded and industrial environments;
- Distributed Ledgers: use of decentralized technologies for secure auditing and “notarized authentication” of data flows generated by sensors;
- Secure Networking: design of secure network architectures based on industrial messaging protocols and encrypted transport channels.

Technologies used: Python, MQTT (Mosquitto), mTLS / OpenSSL, Blockchain, Docker, cryptography libraries.

Link <https://github.com/giumatt/Smart-MicroGrid>

● TRAINING EXPERIENCE

07/2025

UNIMED StartCUP 2025

Active participation in the preparatory training programme ‘StartCUP Competition 2025’ held at the University of Calabria on 8 July 2025, promoted by UNIMED, ETF, AUF and HOMERe, with the participation of international experts in the field of innovation and entrepreneurship.

I strengthened the following skills:

- In-depth study of key entrepreneurial skills through practical exercises and interaction with industry experts, including the European Training Foundation and the Lebanese University;
- Activities focusing on innovative technologies and artificial intelligence applied to entrepreneurship, with the participation of IT experts from the international network and skill experts;
- Collaboration and networking in a multicultural environment, preparation for the StartCUP 2025 competition and discussion with a community of educators and innovation professionals.

This experience strengthened cross-cutting skills such as problem solving, teamwork and understanding of startup dynamics in an international context.

● CERTIFICATIONS

10/06/2016 – CURRENT

Trinity GESE Grade 5 English Language Certification

English language certificate at *level B1* and *GESE* (Graded Examination in Spoken English) issued by **Trinity College London**.

13/12/2016 – CURRENT

Base ECDL Certification

European Computer Driving License certification issued by **AICA** (Italian Association for Information Technology and Automatic Calculation).

European Computer Driving Licence certification issued by **AICA** (Italian Association for Information Technology and Automatic Calculation).

European Computer Driving License certification issued by b (Italian Association for Information Technology and Automatic Calculation) and accredited by **Accredia** (Italian accreditation body).

European Computer Driving Licence certification issued by **AICA** (Italian Association for Information Technology and Automatic Calculation) in the field of IT security.

● **DRIVING LICENCE**

Driving Licence: AM

Driving Licence: B

● **COMMUNICATION AND INTERPERSONAL SKILLS**

Communication and interpersonal skills

Good command of written and verbal communication, listening and understanding the needs of others.

I have a good aptitude for listening and interpersonal communication, which are useful for understanding the needs of others, as well as excellent writing skills. My communication and interpersonal skills were developed both at university and outside of it: my experience playing rugby allowed me to improve my ability to work within a team, collaborate towards a common goal, and respect roles and responsibilities.

● **ORGANIZATIONAL SKILLS**

Organizational skills

Good problem-solving skills and ability to work well in a team or as part of a staff.

I authorize the processing of my personal data contained in my CV pursuant to Article 13 of Legislative Decree No. 196 of 30 June 2003 - 'Personal Data Protection Code' and Article 13 of GDPR 679/16 - 'European Regulation on the protection of personal data'.

Rende (CS) , 04/08/2026



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